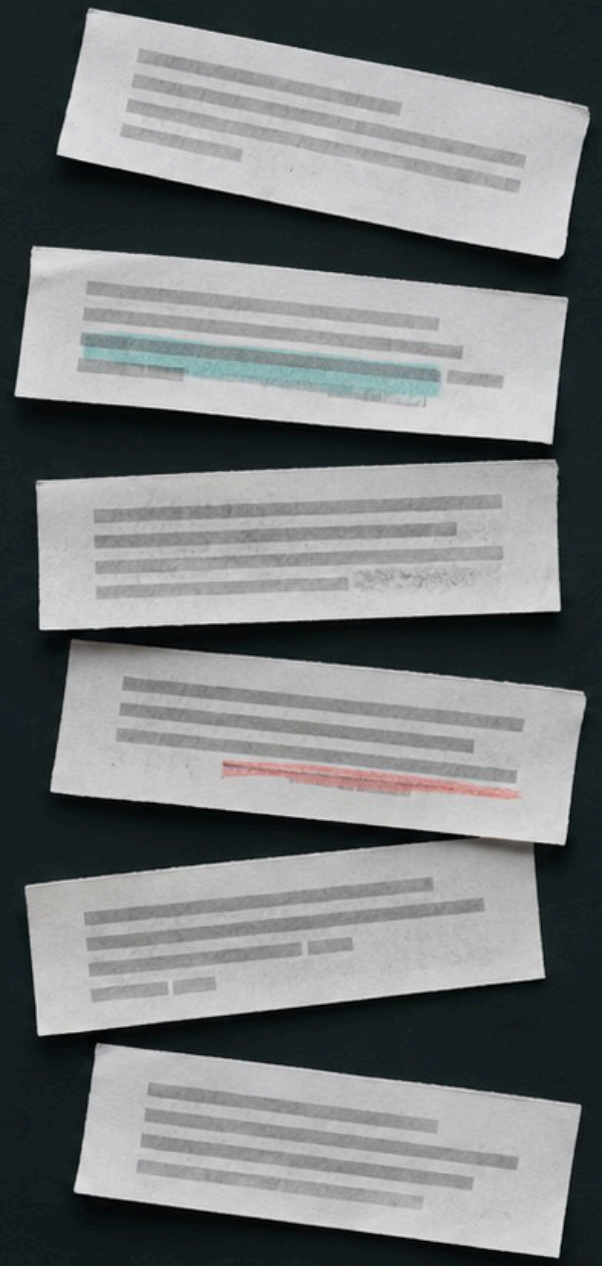
DAMDID 2026

Deep Re-annotation and Controlled Candidate Pools

Re-evaluating ranking models on MS MARCO



A relevant passage can receive zero gain

“what causes boiling point to increase”

Passage (shortened paraphrase)	Original qrels	Human judgment
7112148 Adding salt raises the boiling point.	Relevant	Relevant
7399382 A non-volatile solute raises the boiling point.	Unjudged	**Relevant**
987600 The boiling temperature of acetone.	Unjudged	Not relevant

qrels: query relevance judgments. Unjudged passages receive zero gain.

Human judgments and a larger pool experiment

English MS MARCO

8.84 million passages

Human judgments

1,179

queries; 33,667 pairs

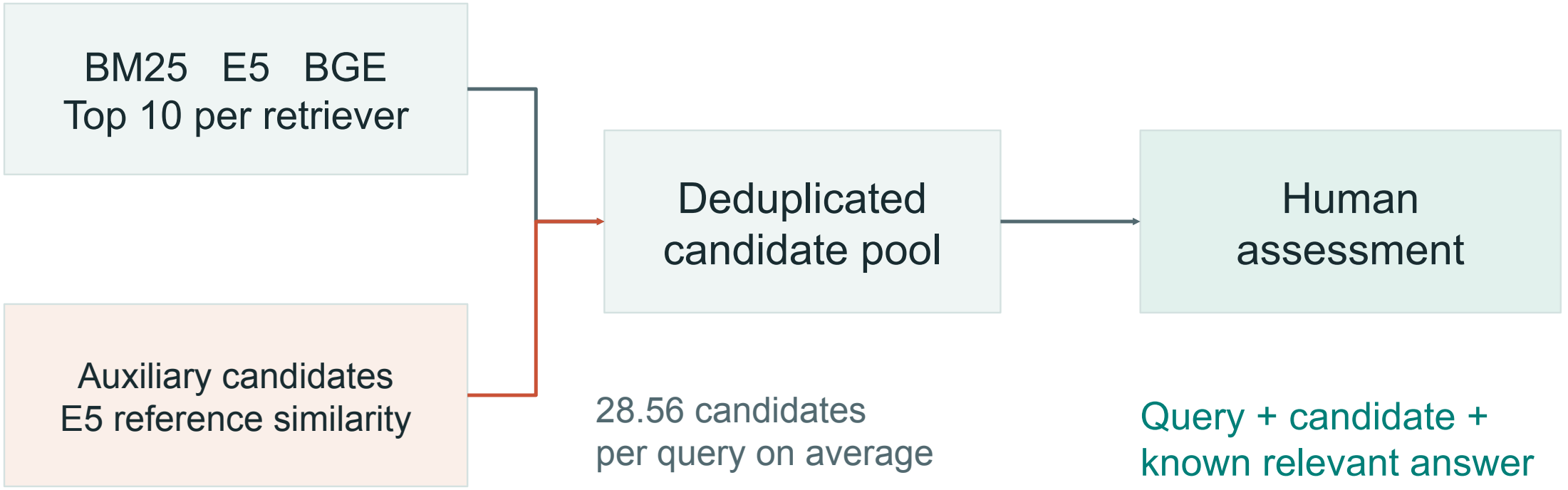
Judge predictions

101,093

queries; varying pool depths

“Deep” describes judgment coverage within selected candidate pools.

Reference-assisted manual annotation



Reference information supports candidate preparation and human assessment.

More relevant passages within the same pools

Relevant judgments per query

1.05

17.47

Original qrels

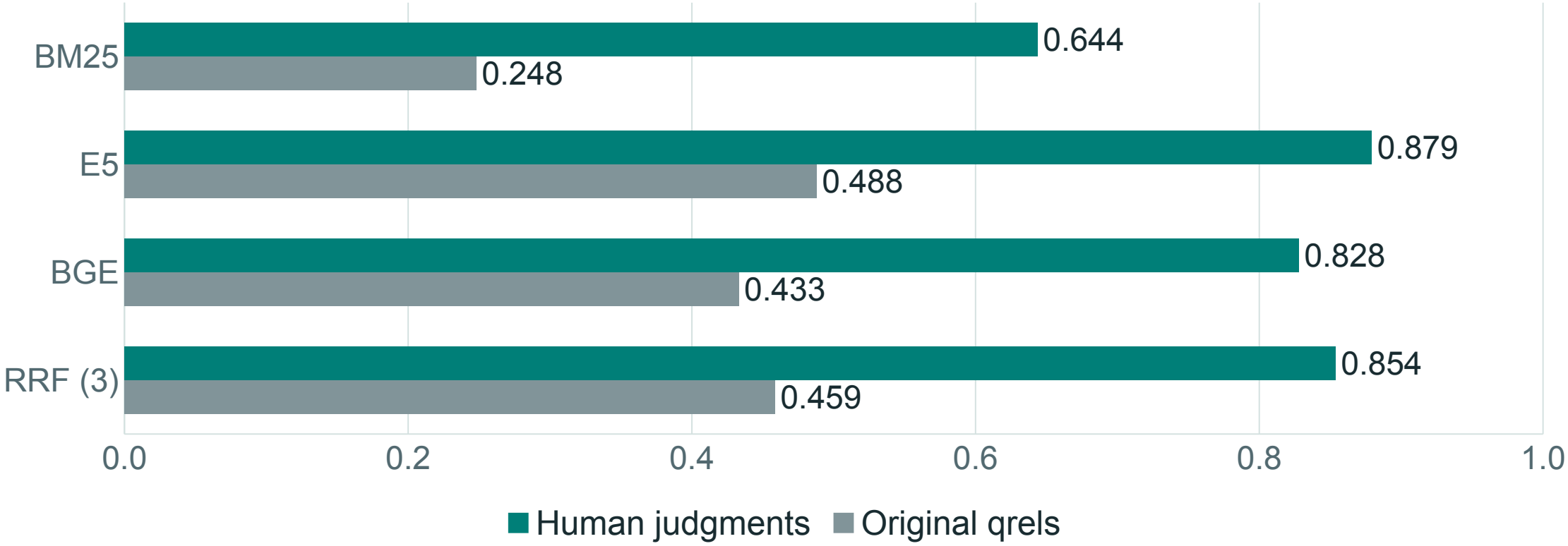
1,243 positive pairs

Extended human judgments

20,597 positive pairs

1,179 queries. Same candidate pools and retrieval rankings.

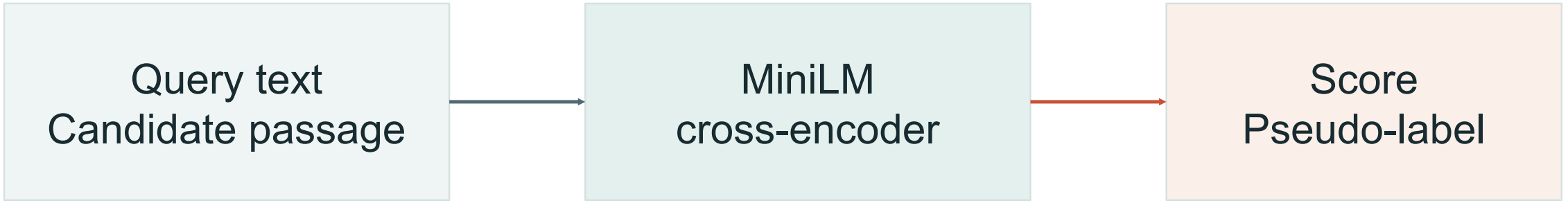
Denser judgments change measured NDCG



E5 paired change: +0.391 95% CI [0.365, 0.417]

Pooled NDCG@10. Same rankings; shared ideal denominator within each query and label set.

A compact judge extends label coverage

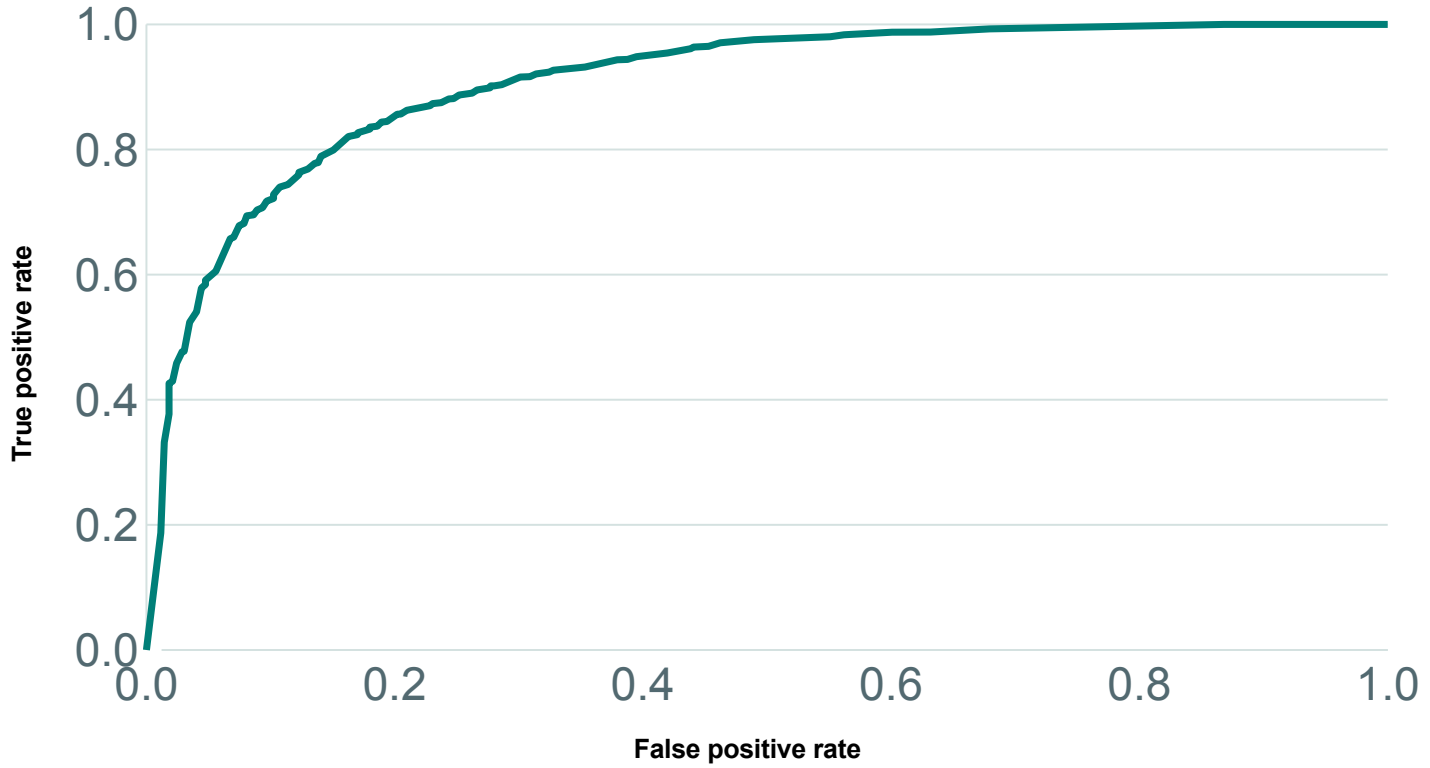


Jointly scores the two texts

Inputs exclude the reference answer, ranks and retrieval scores.

Pseudo-labels are model predictions. The original retrieval rankings remain available.

Judge agreement on held-out queries



0.907

ROC-AUC

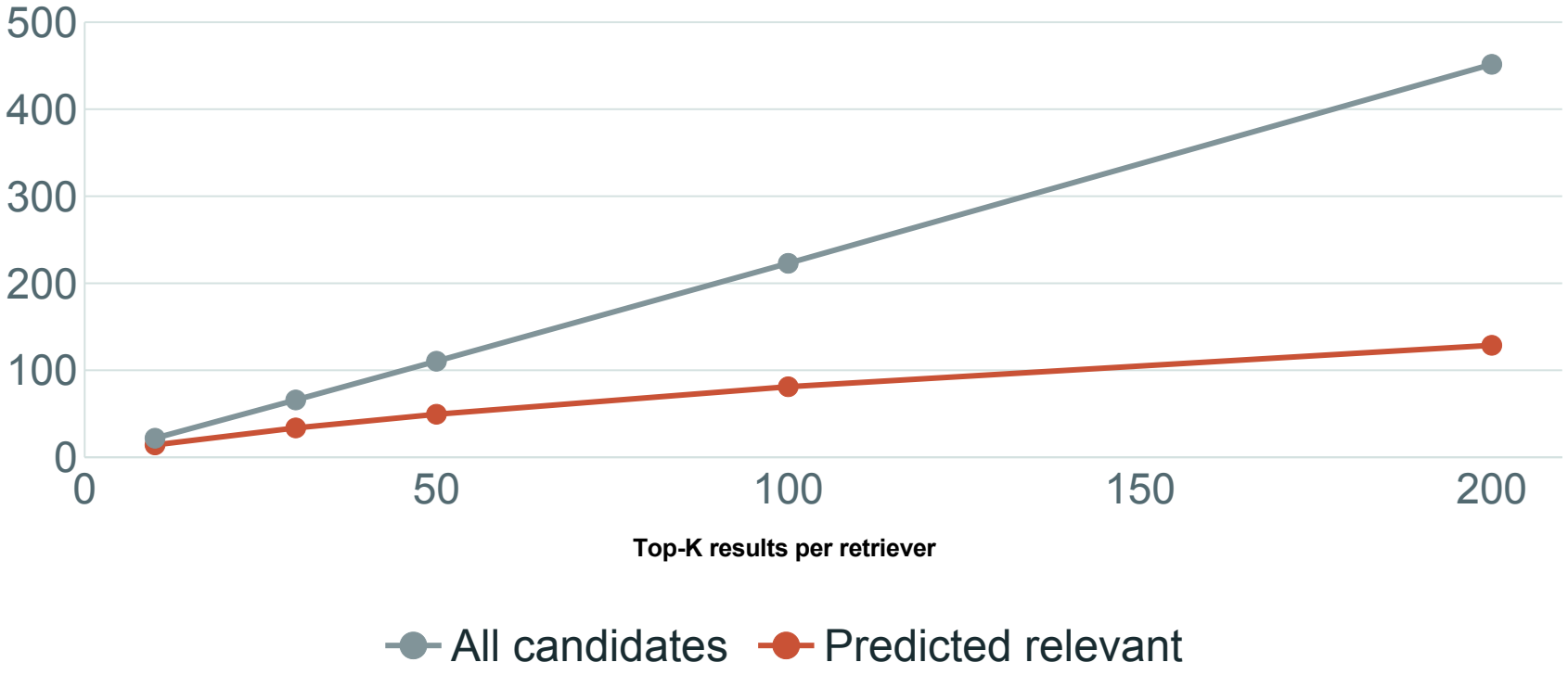
0.875

F1 at validation threshold

177 held-out queries; 5,002 pairs. Inherited binary label encoding.

Larger pools contain more predicted positives

Mean passages per query



451.5

candidates

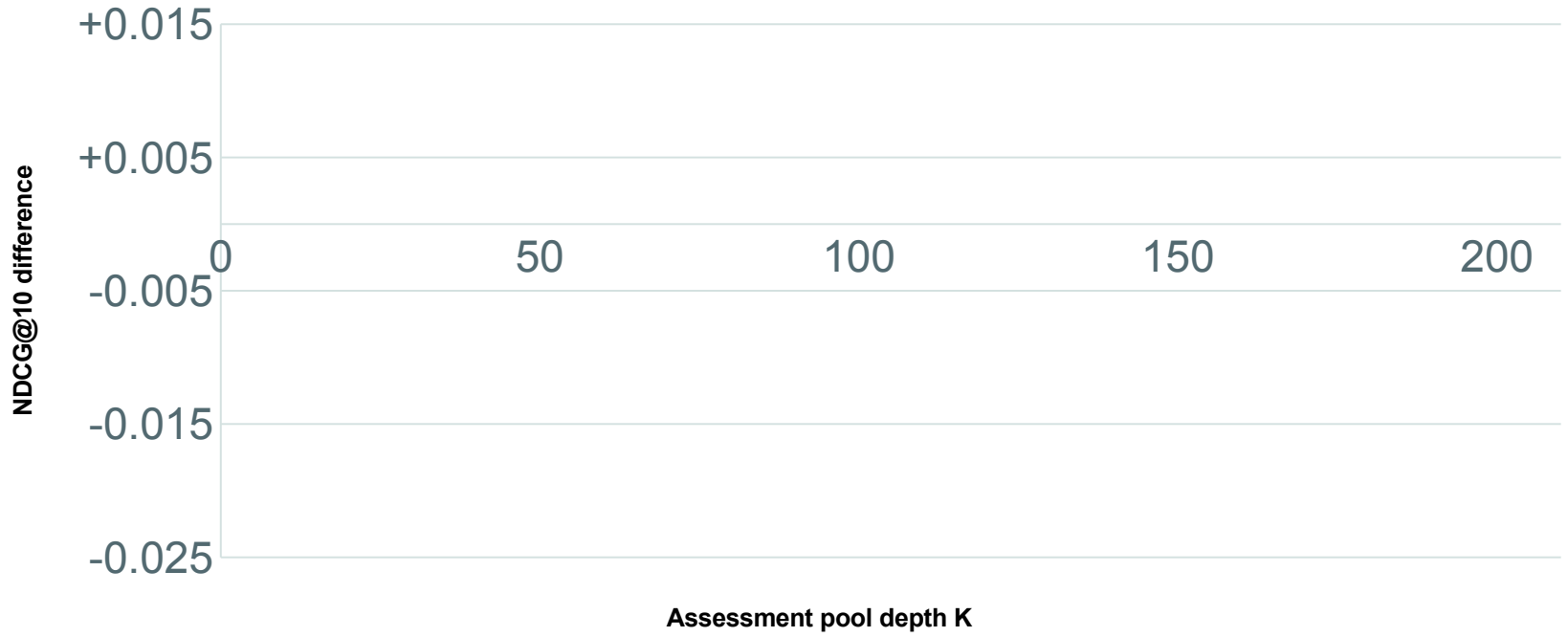
128.6

predicted relevant
at K = 200

101,093 queries, including zero-positive pools. Same judge and threshold.

RRF changes position relative to E5

Assessment depth K = RRF input depth L



E5 higher

−0.0195

K = 10

RRF higher

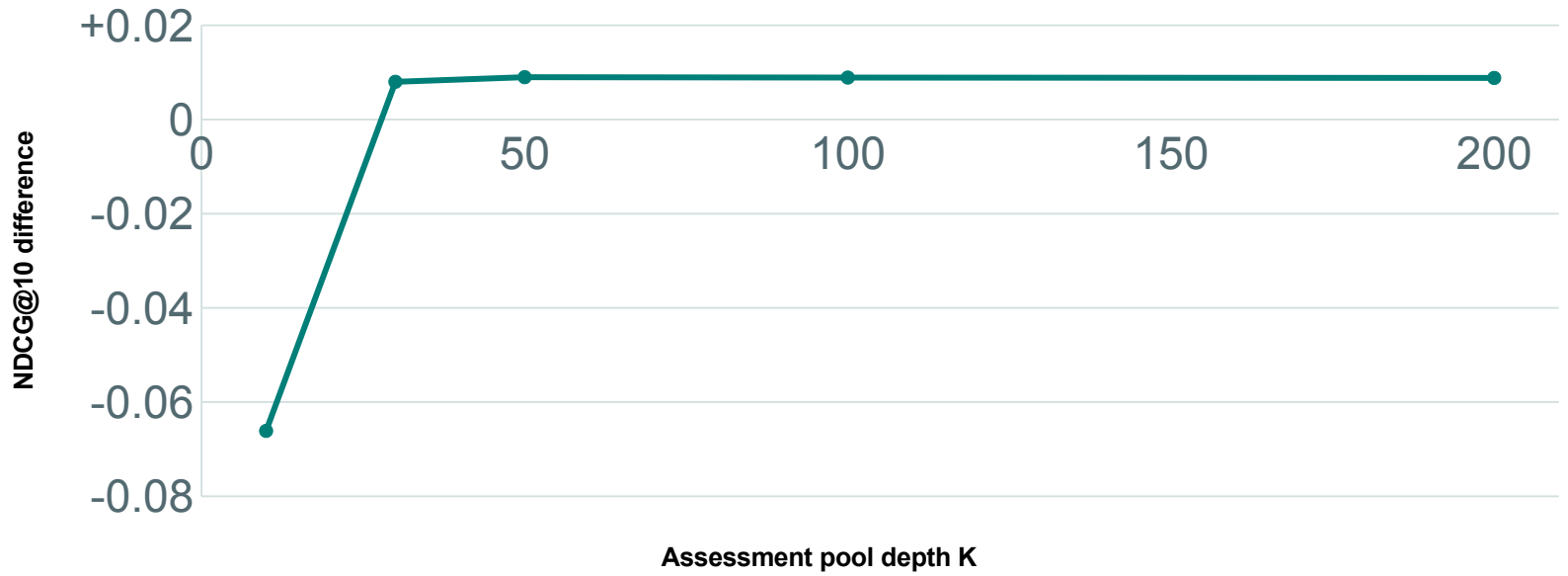
+0.0088

K = 200

Single-judge pseudo-labels. Points with 95% paired query-bootstrap intervals.

Assessment coverage can reverse a comparison

Fixed RRF inputs: top 200. Both output rankings stay unchanged.



14.4%

of returned RRF pairs outside the pool at K = 10

145,643 query–passage pairs

RRF minus E5 under pseudo-labels. Unjudged returned passages receive zero gain.

Scope and robustness

Sensitivity checks

Evaluation	Remaining queries	Observed result
Manual labels 23 affected queries excluded	1,156	System order unchanged
Full-set pseudo-labels Manual queries excluded	99,914	Comparison directions unchanged

Inter-assessor agreement is unmeasured.
Full-set results lack independent human validation.

Evidence concerns selected English MS MARCO pools and one judge.

Judgment coverage affects retrieval evaluation

Human judgments

E5: 0.488 to 0.879

Same ranking; more complete relevance judgments

Fixed RRF and E5 rankings

Assessment coverage alone can change their order

English MS MARCO; human judgments and single-judge pseudo-labels.

NDCG@10 and paired annotation effects

NDCG@10 = DCG@10 / ideal DCG@10

Ranking	Human − original	95% paired CI
BM25	+0.396	0.372 to 0.419
E5	+0.391	0.365 to 0.417
BGE	+0.395	0.369 to 0.419
RRF (3)	+0.395	0.371 to 0.419

5,000 paired query bootstraps. All four annotation effects: $p < 0.001$.

Judge test protocol and observed errors

825 / 177 / 177 queries

Train / validation / test

Test target	Predicted negative	Predicted positive
Negative under inherited mapping	**1,300**	506
Positive	314	**2,882**

Precision 0.851 Recall 0.902 F1 0.875

5,002 pairs; six unresolved targets mapped to zero and one empty-text record.

Full-set depth results

K	Candidates / query	Predicted relevant / query	E5 NDCG	RRF L=K NDCG	RRF L=200 NDCG
10	21.9	14.2	0.871	0.852	0.805
30	65.8	33.7	0.851	0.862	0.859
50	110.4	49.4	0.847	0.856	0.856
100	222.9	81.0	0.844	0.851	0.853
200	451.5	128.6	0.842	0.850	0.850

Single-judge pseudo-labels. All 101,093 queries remain in every average.

Sensitivity to query exclusions

Ranking	All 1,179 queries	Clean 1,156 queries
BM25	0.644	0.644
E5	0.879	0.879
BGE	0.828	0.828
RRF (3)	0.854	0.855

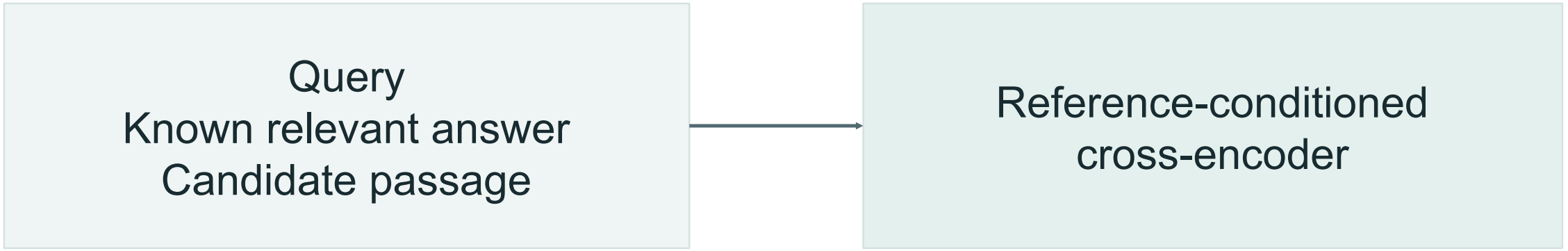
Full set without manual queries: N = 99,914

At K = 200: RRF − E5 = +0.00877

Exclusion checks preserve the observed directions, not independent human validity.

Next validation: reference-conditioned judging

Proposed input



Compare stronger judges against independent human assessments.

Planned evaluation. No results for this model are reported in the present study.